



Network Swiss Army Knife (NSAK)

Bachelor Thesis – Integrating Agentic AI for Network Reconnaissance

June 12, 2026

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Introduction

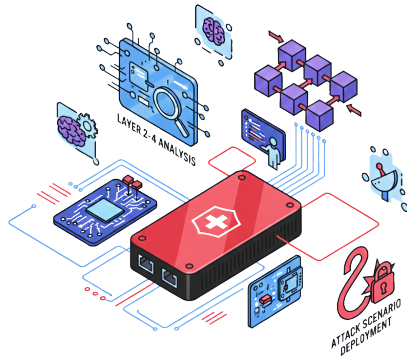
Starting Point: The NSAK Framework

Established in the predecessor project (Project 2)

A modular, open-source network security framework

Built around four reusable resource types:

- 🖨️ *Device execution host at a network position*
- 🏗️ *Environment target network topology*
- 📁 *Scenario a red/blue team use case*
- ⚙️ *Drill reusable building block*



Hypothesis and Research Questions

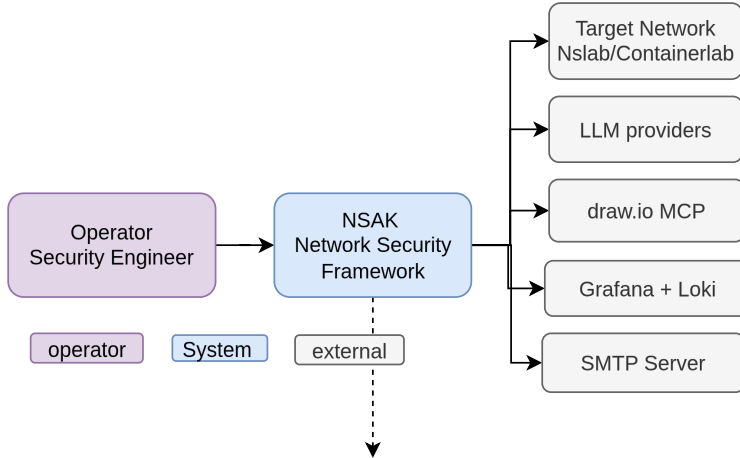
From static scenarios to agentic AI

Hypothesis

Integrating agentic AI into NSAK improves adaptability, flexibility and automation over static scenarios, while lowering the operator's required expertise to interpret results and plan follow-up steps.

Layered Architecture

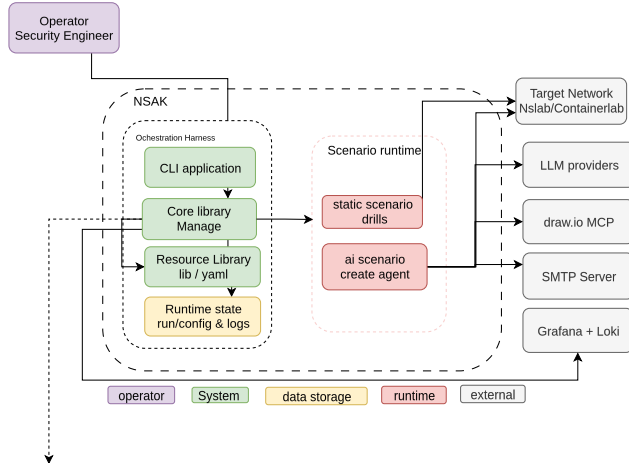
system-diagram



Background and Concepts

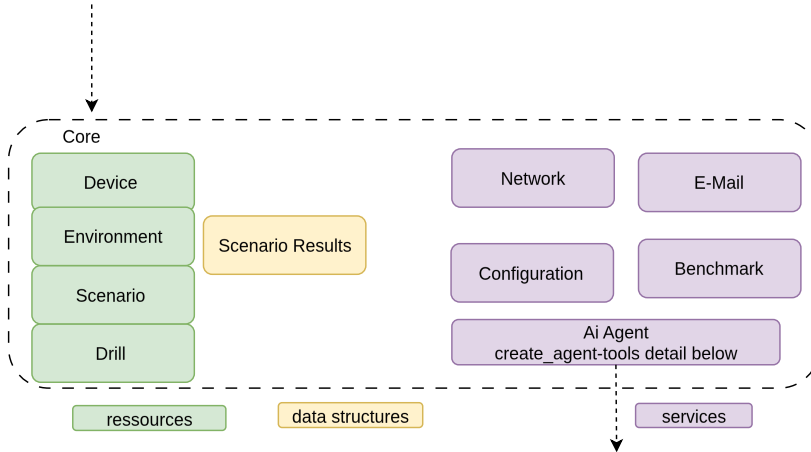
Layered Architecture

Lean Component Diagram



Layered Architecture

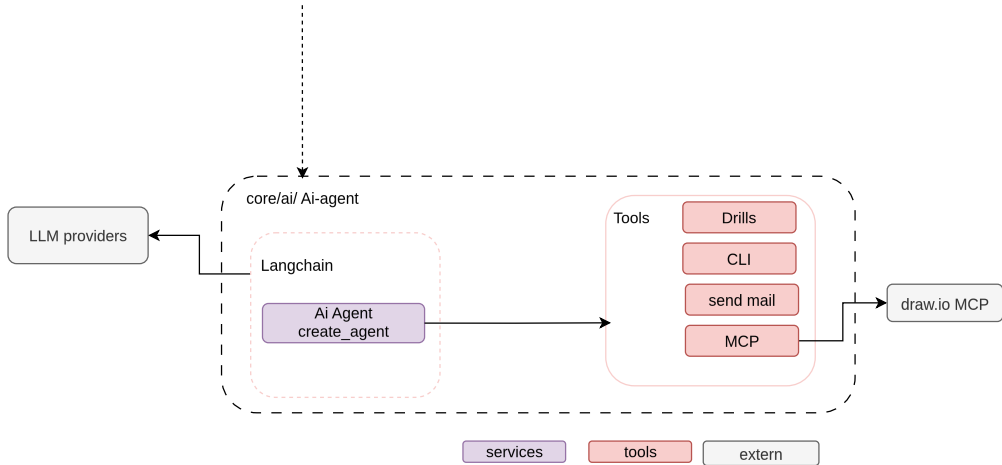
Core Diagram and Agent



Implementation

Layered Architecture

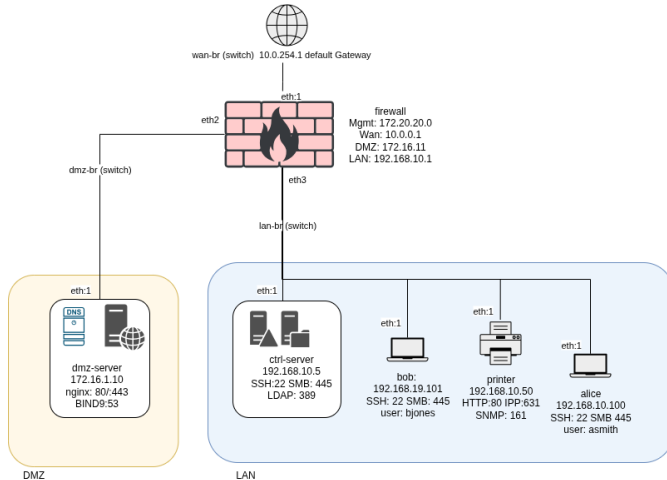
Core Diagram and Agent



Evaluation Setup

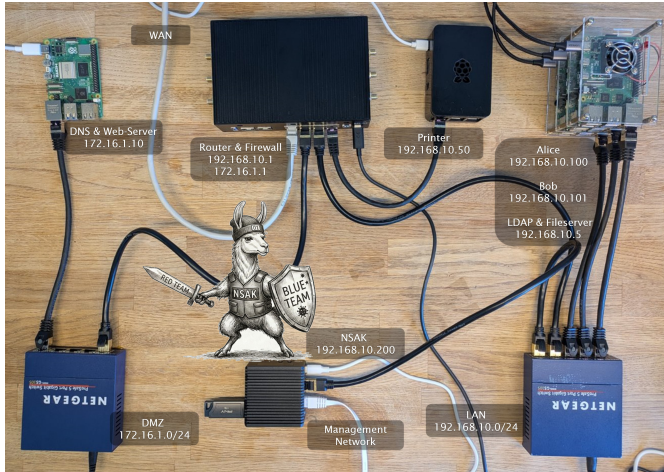
Reproducible Test Environment

A virtual enterprise network built with Containerlab



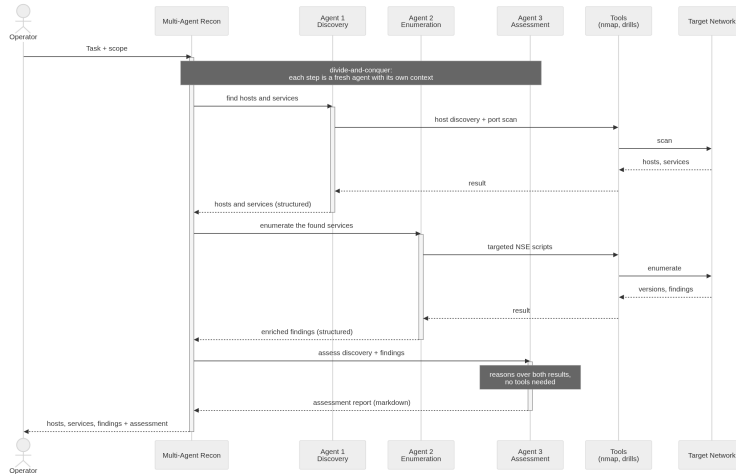
Physical Lab

A virtual enterprise network built with Containerlab



Multi-Agent Reconnaissance Pipeline

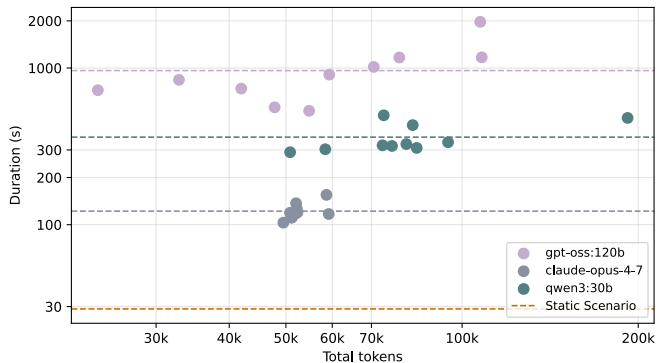
Discovery → enumeration → assessment, each a fresh agent



Results

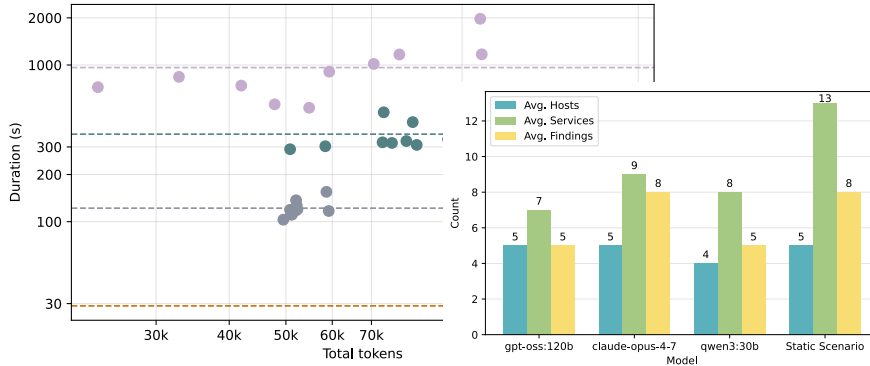
Containerlab – Multi-Agent Benchmark

Containerlab Findings



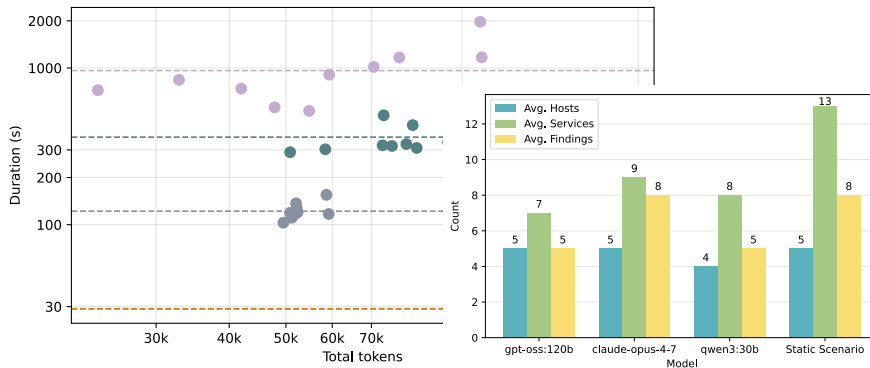
Containerlab – Multi-Agent Benchmark

Containerlab Findings



Containerlab – Multi-Agent Benchmark

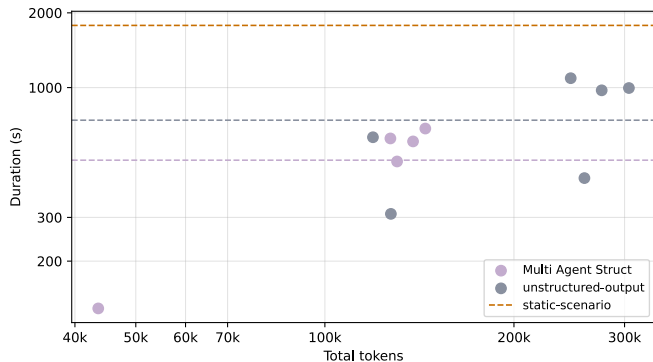
Containerlab Findings



- Decomposition: **qwen3:30b** drops tokens –68% and duration –30%.
- **claude-opus-4-7** keeps identical discovery quality (slight +13% token overhead).

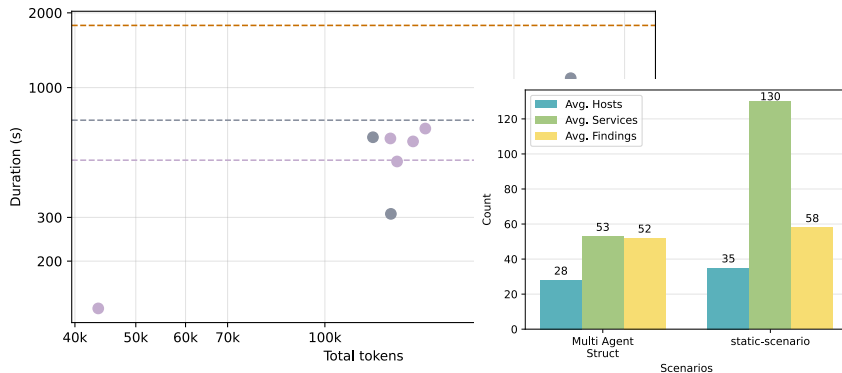
Network Security Lab – Multi-Agent Benchmark

NSLab - Findings



Network Security Lab – Multi-Agent Benchmark

NSLab - Findings



Discussion and Conclusion

Key Findings

Takeaways across both environments

✓ What works

- Even smaller models can autonomously run **standard red-team reconnaissance**
- **Multi-agent decomposition** helps smaller models (-68% tokens for **qwen3:30b**)
- AI delivers **depth**: assessment, prioritized findings, remediation

Key Findings

Takeaways across both environments

✓ What works

- Even smaller models can autonomously run standard red-team reconnaissance
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⚠ What limits it

- High failure rate – best-case subset
- Hallucinations: fabricated CVEs, honeypots

Recommendations and Future Work

Recommendations

- Match model tier to task – frontier model for autonomous recon
- Multi-agent for smaller / self-hosted models
- Hybrid pipeline: static breadth + AI depth (+ RAG)
- Human-in-the-loop + retry on incomplete runs

Future work

- Dynamic tool calling – expose only relevant tools
- Guardrails & sandboxing – scope restriction for production
- Robust HITL middleware
- Broader attack-chain coverage beyond reconnaissance

Demo: Agentic Reconnaissance in Action

Questions?